

MOHAMMED REZEQ AWAD

Computer Engineering | AI & Machine Learning | Data Analysis | Software Development

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PROFESSIONAL SUMMARY

Computer Engineering specializing in Intelligent Systems, with hands-on experience building end-to-end machine learning pipelines, frontend web applications, and mobile apps with Flutter. Currently leading a development team building an AI-powered media workflow platform. Comfortable across the full workflow, from preparing real datasets to tuning models and shipping working software, with a strong foundation in data structures and object-oriented design. Learns fast, builds consistently, and turns ideas into products that solve real problems.

EDUCATION

B.Sc. in Intelligent Systems & Computer Engineering

Sep 2022 – Jul 2027 (expected)

Al-Aqsa University, Faculty of Engineering, Gaza, Palestine

- Relevant coursework: Artificial Intelligence, Machine Learning, Pattern Recognition, Data Structures & Algorithms, Software Engineering, Control Systems, Advanced Circuits, Computer Networks, Cybersecurity.

General Secondary Certificate, Scientific Stream

2022

GPA: 89.4

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, Dart

AI / Machine Learning: scikit-learn, XGBoost, TensorFlow/Keras, pandas, NumPy, Matplotlib, LangChain, Jupyter/Colab

ML Techniques: Classification, feature engineering, class imbalance handling, hyperparameter tuning, cross-validation, model evaluation (ROC AUC, F1, precision/recall)

Web Development: HTML5, CSS3, JavaScript, responsive design, Arabic/English RTL internationalization

Mobile Development: Flutter, SQLite, SharedPreferences

Engineering Tools: MATLAB, Cisco Packet Tracer, Delta WPLSoft (PLC), Git & GitHub, VS Code

Core Concepts: Object-oriented programming, data structures, algorithms, database design, circuit analysis

EXPERIENCE

Team Leader, Taqat Academy

Jul 2026 – Present

Gaza, Palestine

- Lead a development team building Nashr AI, an AI media workflow platform that takes field reporting from raw material to a publication-ready package.
- Own product scoping and technical architecture across four areas: Flutter mobile app, Laravel backend, Python/FastAPI AI services, and React editor dashboard.
- Coordinate the team through a 12-week build cycle and present the product concept and progress to mentors, including at the program's ideas showcase.

Freelance AI Specialist

2024 – Present

Self-employed

- Produce AI-generated visual concepts for interior design, used by clients for decoration in homes, cafés, halls, and event spaces.
- Translate client briefs into precise prompts, iterating on style, lighting, and layout to deliver usable design references.

SELECTED PROJECTS

Bank Customer Churn Prediction, Machine Learning

2026

- Built an end-to-end ML pipeline on a 10,000-customer dataset (~20% churn) to predict which customers are likely to leave.
- Engineered features with one-hot encoding and standard scaling, using a stratified, leakage-free train/validation/test split.
- Addressed class imbalance through class weighting to prioritize recall, then trained and compared Logistic Regression, Random Forest, XGBoost, and a Keras neural network.
- Tuned XGBoost with randomized search and 5-fold cross-validation, achieving ~0.86 ROC AUC and ~0.72 recall on unseen test data.

Gaza Aid Portal, Full-Stack Web Application

2025

- Developed a bilingual (English/Arabic) platform allowing people to register once for humanitarian aid from multiple NGOs and track their request status.
- Implemented one-click language switching that translates the full interface and flips layout between LTR and RTL, with the preference persisted across pages.
- Built registration, status lookup, complaint reporting, and support modules with client-side and server-side validation.

Pattern Recognition Study, "When Feature Scaling Changes the Optimal Classifier"

2026

- Investigated how preprocessing can reorder classifier performance, combining mathematical analysis with controlled experiments.
- Demonstrated a ranking inversion: the best model on raw data was scale-invariant (~ 0.95), while after scaling SVM and k-NN rose from ~ 0.70 to ~ 0.98 accuracy.

Library Management System, Java and Custom Data Structures

2025

- Implemented generic stack, queue, linked list, and dynamic array from scratch instead of using Java's built-in collections.
- Applied each structure to its natural use case: a FIFO queue for pending borrow requests, stacks for per-book and per-user borrowing history.
- Built role-based access for students, employees, and librarians, with authentication, account status checks, and transaction validation.

IT Company Management System, Java and OOP

2023

- Designed a class hierarchy using inheritance, abstraction, and polymorphism to manage managers, engineers, customers, and projects.
- Implemented role-specific salary calculation through an overridden abstract method, and object serialization for persistent storage.

Machine Learning Research Paper, Al-Aqsa University

2026

- Co-authored a paper surveying ML paradigms, landmark architectures, cross-industry applications, and AI ethics (fairness, privacy, transparency, accountability), plus the 2026 shift toward efficient specialized architectures, Explainable AI, and Edge AI.

Additional Projects

- Flutter mobile apps: a Smart University Portal and a Student Management System with local SQLite persistence.
- AC filter circuit analysis of a 2-way loudspeaker crossover (transfer functions, ~ 3.09 kHz cutoff, quality factor); industrial automation programs in Delta WPLSoft; university network design in Cisco Packet Tracer.

CERTIFICATIONS

- Google AI Professional Certificate, Google / Coursera, 2026 (7 courses: AI Fundamentals, Brainstorming & Planning, Research & Insights, Writing & Communicating, Content Creation, Data Analysis, App Building)
- Google AI Essentials Specialization (5 courses) and Introduction to Responsible AI, Google / Google Cloud / Coursera, 2026
- Anthropic, 2026: Claude with the Anthropic API; Claude Code 101; AI Fluency: Framework & Foundations; AI Fluency for Students
- LangChain Academy, 2026: Introduction to LangChain (Python); LangChain Essentials (Python)
- Introduction to Data Analytics, IBM / Coursera, 2026
- English Language Program, Level 12 (Advanced), AMIDEAST

LANGUAGES

Arabic	Native
English	Advanced (completed all 12 levels at AMIDEAST)

ACTIVITIES & LEADERSHIP

- Student Member, IEEE (Institute of Electrical and Electronics Engineers), Palestine Subsection, 2026.
- Team leader on university engineering projects; regularly tutor peers, simplifying complex technical topics before exams.